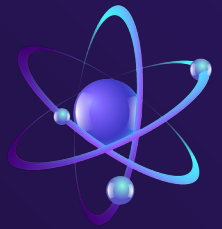


PERSONALIZED MEDICINE AND CELL-BASED THERAPIES

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02

CELL BASED THERAPIES

Esha Zaman

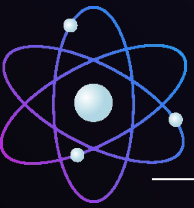


What if we could reprogram a patient's own immune cells to find and destroy their cancer with amazing precision?



THIS IS NOT JUST A DREAM!!

CAR T CELLS THERAPY



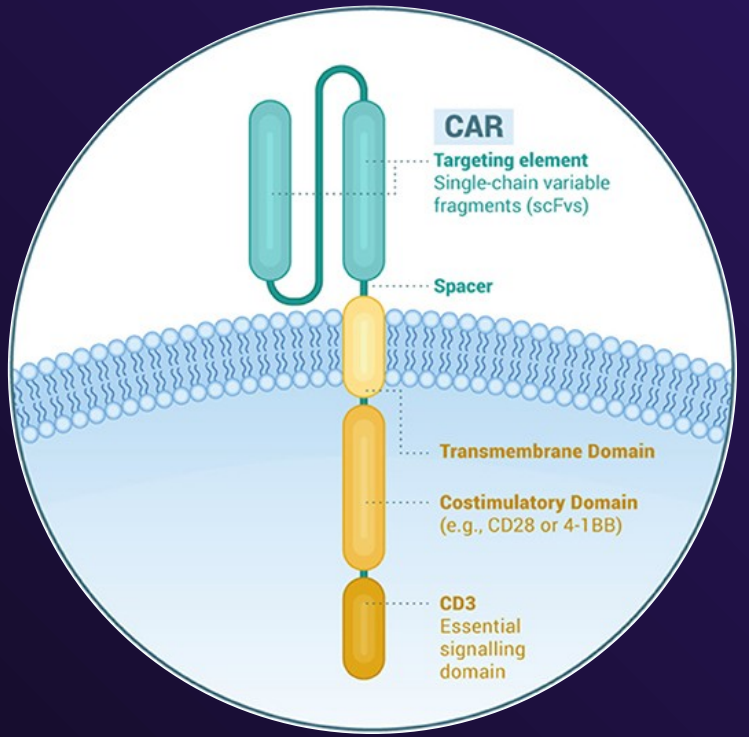
CAR T CELLS THERAPY

1. CAR T cell therapy uses a patient's own immune cells to target and destroy cancer cells

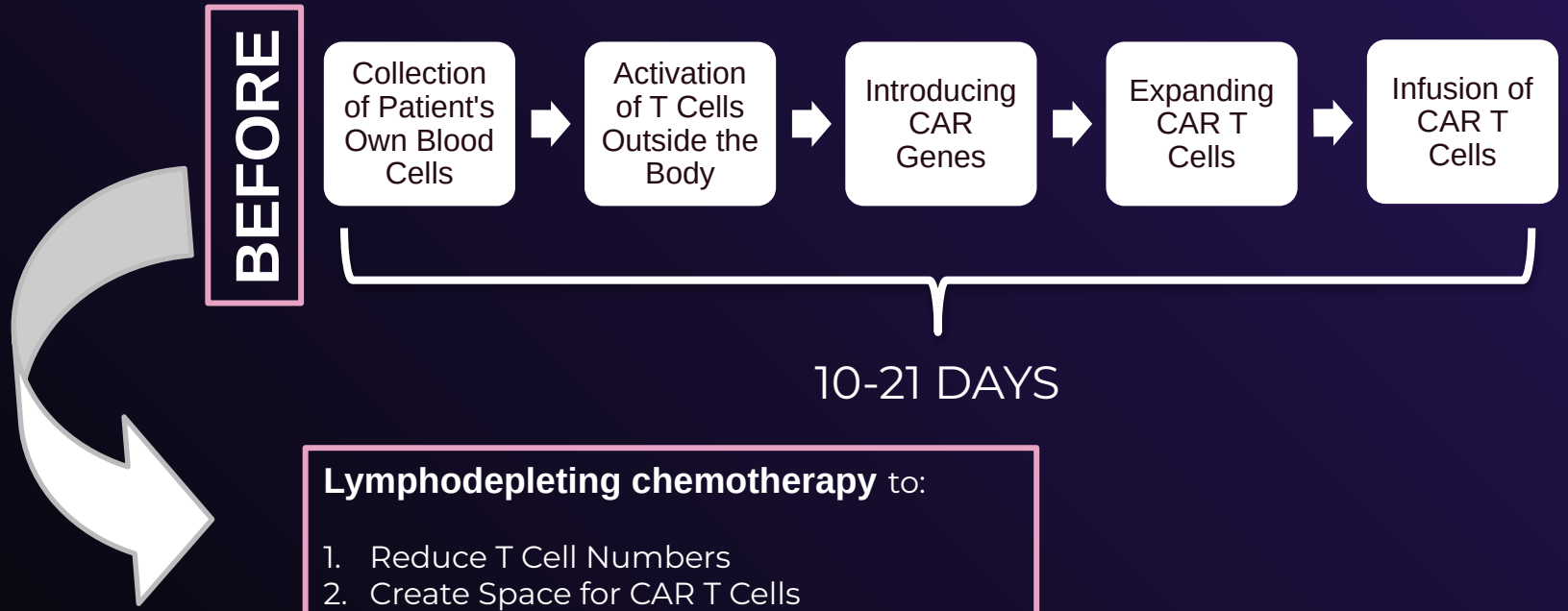
2. CAR stands for **Chimeric Antigen Receptor**, a specially engineered receptor that is added to T cells to help them better recognize and attack cancer cells

3. Simplistically, a CAR is comprised of an:
   - Extracellular domain derived from antibody
   - Short linker between heavy and light chain
   - Transmembrane domain
   - Intracellular T cell signaling domain

4. Second- and third-generation constructs have one or two costimulatory signaling domains



ADMINISTRATION OF CAR T CELLS





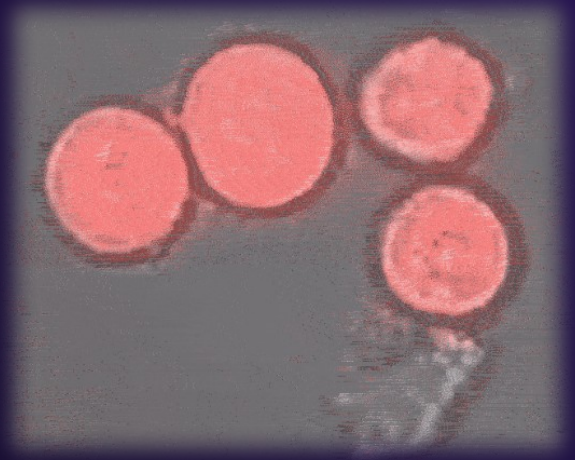


FIRST PEDIATRIC PATIENT TO RECEIVE CAR T-CELL THERAPY

- Emily's CAR T journey began in march 2012
- It has been 12 years since Emily was declared cancer-free

LIMITATIONS

1. Difficulty in producing CAR T cells for every patient who needs them
2. Relapse remains a concern, often occurring within a year of treatment
3. Neurotoxicity
4. B cell aplasia: destruction of normal B cells in addition to cancerous ones
5. Cytokine release syndrome



Did you know that over 50% of organ transplants fail due to immune rejection?

“Imagine if we could eliminate this risk entirely.”



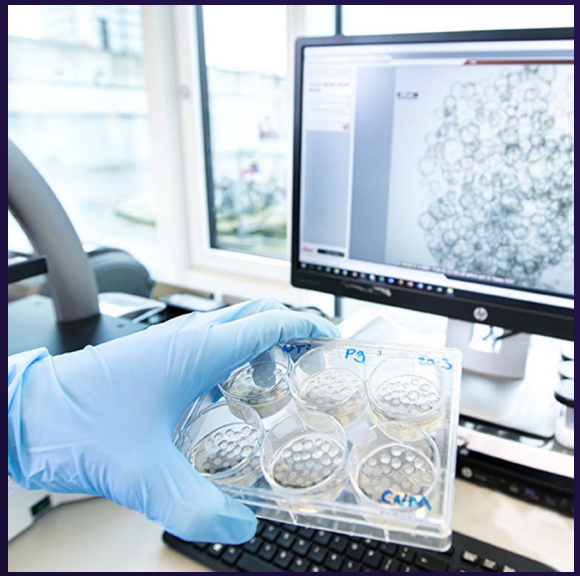
AUTOLOGOUS STEM CELL THERAPIES

AUTOLOGOUS STEM CELL THERAPIES

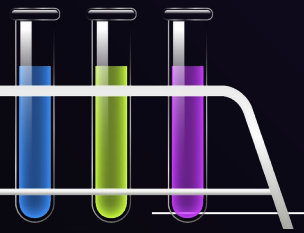
1. Stem cells are unique cells with the ability to renew themselves and differentiate into various cell types.
2. Autologous stem cell therapy, uses a patient's own cells, significantly reducing the risk of rejection and improving treatment outcomes.

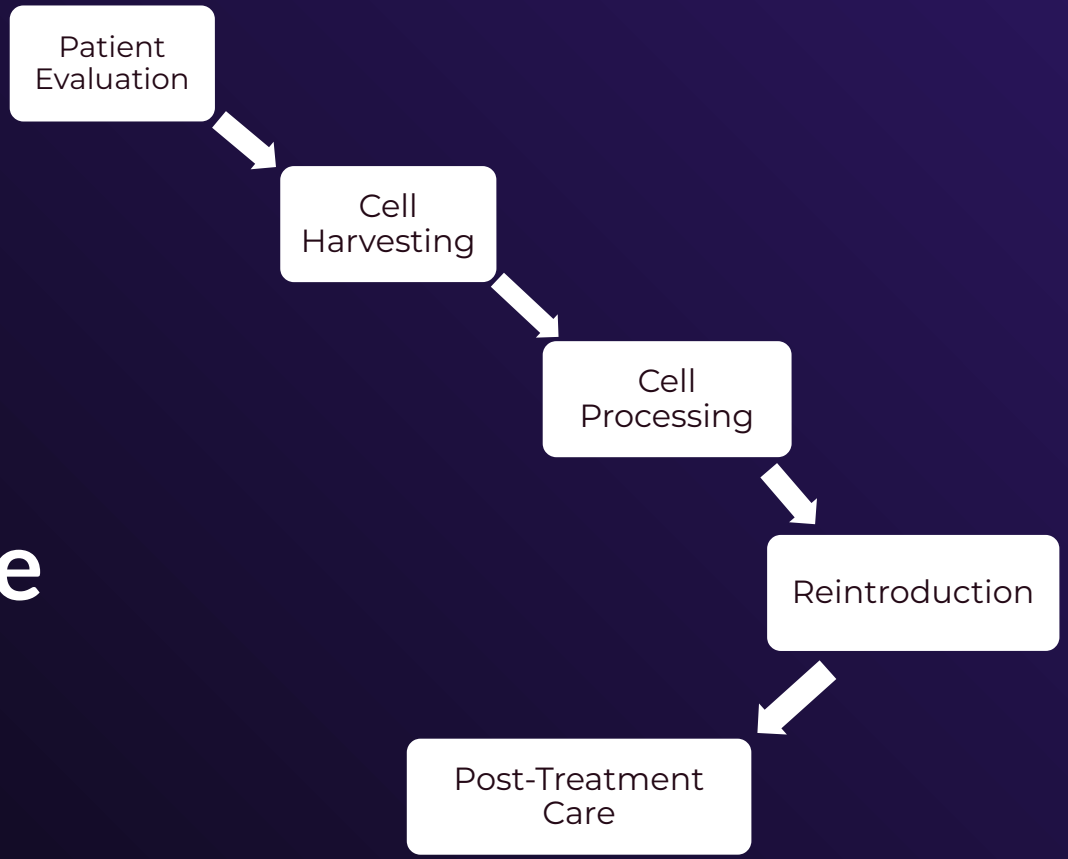
3. Advantages of Autologous Stem Cell Therapy are:
- Reduced Risk of Rejection
- Personalized Treatment
- Fewer Side Effects

4. Can be used for:
- Orthopedic Injuries
- Hematological Conditions
- Cardiovascular Diseases



Autologous Stem Cell Therapy for Knee Cartilage Repair





THANKS

DO YOU HAVE **ANY QUESTIONS?**

